# CONTROLADOR DE VELOCIDAD LAZO CERRADO TDA1085C — AGUJEREADORA DE BANCO (ACOPLE DIRECTO)
# Motor: Askoll C.E.SET. MCA 52/64-148/ALL7 (220V 50Hz, 330W, Tacometro 60-180R)
# Triac: BTB12-600BW (I_GT=50mA, ALETA VIVA NO AISLADA CONECTADA A MT2 — AISLAR DEL CHASIS)
# Rango: 400 a 2.500 RPM (Preset 50k multivuelta RV1 en serie con Potenciometro 10k de panel)
# Star Ground: L1 entra a MT1 de Q1 y sale en pista unica hacia Pin 8 (GND_CHIP)

J1
GND_CHIP 1
MT2 2
L2_AC_HOT 3
L2_AC_HOT 4
L1_MT1
MT2_MOT7
L2_MOT6
L2_AC_HOT
AC_220V_A_MOTOR

Q1
GND_CHIP 1
MT2 2
MT1
GATE
MT2_TAB
3 GATE
BTB12-600BW (NO AISLADO)

C_SNUB1
MT2 N_SNUB
100R 1W
N_SNUB GND_CHIP
100nF 400V X2

R_GATE1
PIN13_TRIG GATE
150R 1W

D1
L2_AC_HOT N_RECT
1N4007

R_DROP1
N_RECT N_DROP_MID
4.7k 3W (Elevar 5mm)

R_DROP2
N_DROP_MID VCC
4.7k 3W (Elevar 5mm)

C_VCC1
VCC GND_CHIP
220uF / 25V

C_DEC1
VCC GND_CHIP
100nF

R1
MT2 PIN1_ISYNC
820kR 1/2W

R2
L2_AC_HOT PIN2_VSYNC
820kR 1/2W

R3
PIN3_ILIM GND_CHIP
10k

C1
PIN3_ILIM GND_CHIP
100nF

RV1
VCC 1
N_POT_TOP 3
2 N_POT_TOP
50k Multivuelta

J2
N_POT_TOP 1
PIN5_VSET 2
N_POT_BOT 3
TOP
WIPER
BOT
POT_PANEL_10k

R_MIN1
N_POT_BOT GND_CHIP
2.2k (Piso 5400 RPM)

C2
PIN5_VSET GND_CHIP
100nF

J3
TACHO_AC 1
GND_CHIP 2
Pin_1
Pin_2
TACHO_MOTOR (Pines 3-4)

R12
TACHO_AC PIN12_TACHO
10R

C12
PIN12_TACHO GND_CHIP
10nF (Ajust. 1 Picos)

R11
PIN11_PUMP GND_CHIP
470R 1nF

R10
VCC PIN11_PUMP
47k

R9
PIN4_VTACHO GND_CHIP
270k

C11
PIN4_VTACHO GND_CHIP
470nF

C7
PIN7_RAMP GND_CHIP
4.7uF / 50V

C14
PIN14_SAW GND_CHIP
47nF (100Hz Saw)

R13
VCC PIN15_ISAW
27k

R14
PIN16_COMP N_COMP_INT
47k

C15
N_COMP_INT GND_CHIP
1uF / 50V

C16
PIN16_COMP GND_CHIP
47nF

U1
PIN1_ISYNC 1
PIN2_VSYNC 2
PIN3_ILIM 3
PIN4_VTACHO 4
PIN5_VSET 5
GND_CHIP 6
PIN7_RAMP 7
GND_CHIP 8
I_SYNC LOOP_COMP 16 PIN16_COMP
V_SYNC ISAW_CURR 15 PIN15_ISAW
I_LIM SAW_CAP 14 PIN14_SAW
V_TACHO TRIG_OUT 13 PIN13_TRIG
V_SET TACHO_IN 12 PIN12_TACHO
RAMP_IN PUMP_OUT 11 PIN11_PUMP
RAMP_BIAS VCC 10 VCC
GND_CHIP +VCC 9 VCC

DRC: Clearance >3.0mm HV_AC a GND_CHIP | Star Ground en Pad MT1
Triac BTB12-600BW (No Aislado, I_GT=50mA, R_G=150R 1W, C_VCC=220uF)
Motor Universal Askoll MCA 52/64-148/ALL7 (400 — 2500 RPM)
**UTN — Hardware Libre**

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***Title: Controlador TDA1085C para Agujereadora de Banco (Acople Directo)***

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|---|---|---|
| Size: A3 | Date: 2026-11-11 | **Rev: 2.0** |
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |